

MATH 180 Strategies of Problem Solving

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Outline

- 1 SoPS Lecture 1
 - Welcome

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What is this course about?

Learn what mathematicians do

- play → discover patterns
- abstract → express patterns mathematically
- prove → justify why the pattern is true

Glimpse advanced mathematics

- logic, set theory, cardinality
- probability, combinatorics, graph theory

Make friends

- math is **social**
- you'll see each other *a lot*

Course expectations

Grading

- pre-class assignments →
- in-class assignments →
- homework →
- seminar attendance →
- research project →
- final exam →

Course expectations

Daily

- complete pre-class assignments before class
- attend every class
- participate **actively**
- solicit participation from group members
- attend (at least) three seminars
- research project

Pinching Pennies

Rules

- arrange pennies in two piles
- turn: take one or more pennies *but*
- take only from one pile per turn
- must take at least one penny
- person who takes last penny wins

Activity (20 mins)

- split into pairs
- play 4-5 rounds
- **record data:** initial board, who won?

Pinching Pennies: Debrief

Gather the data

- initial board? **Player 1** or **Player 2** won?
- is it better to be **Player 1** or **Player 2** ?
- if the heaps are equal, who should win?

Challenge the instructor!